Typed Decision Models as Passage Rerankers: Effectiveness, Calibration, and Context Selection

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ABSTRACT

Typed decision models return probabilities over caller-defined options. We evaluate four families as passage rerankers on TREC DL 2019–2023 and eight BEIR collections, and study calibration and retrieval-augmented generation (RAG). Jev averages 56.1 nDCG@10 versus Qwen3-Reranker-8B's 56.0 and bge-reranker-v2-m3's 53.7; pooled superiority and equivalence depend on small-sample inference. Jev's raw yes/no probabilities reduce mean calibration error by five points, although source-label calibration gives bge similar error. Same-base and context controls limit attribution to decision training. RAG uses 1,000 questions per dataset and three generators. Jev improves MuSiQue answers under question-ID splits with substantial component reuse; on 329 questions excluding directly shared components, top-3 point gains persist but none passes the original Holm correction. Probability thresholds establish no gain over fixed top-k.

KEYWORDS

reranking, decision models, calibration, retrieval-augmented generation

1 INTRODUCTION

Rerankers score retrieved candidates pointwise or order them listwise [15, 21]. Typed decision models receive a state and typed questions, returning option probabilities. Whether those probabilities support calibrated context selection beyond competitive rankings remains an empirical question.

We study Laya, Jeff, Clef, and Jev [3–5, 20] without task-specific tuning. We ask: **RQ1**, how effective are they at reranking, including against same-base controls? **RQ2**, are their probabilities calibrated, and can transfer calibration close conventional rerankers' gap? **RQ3**, do probabilities improve downstream context selection? Ranking, calibration, and answer quality require separate evaluation: a good ordering need not support thresholding, and marginal relevance may miss multi-hop utility.

Relation to prior work. Pointwise LLM ranking uses binary or graded token probabilities [22]; our controls test value beyond that readout. Earlier work studies Jev for recommendations [8]; audits caution against interface attribution without comparable controls [17]. Ranker scores can be poorly calibrated [10]. We test probability calibration and its utility against fixed-k and score-gap selection [16].

2 METHODS AND EVALUATION

Typed scores. Each query–passage pair is the model's state. Three questions request: Choice-3 (C3), with relevant, partially relevant, and irrelevant options; Noul, asking whether the passage answers or substantively addresses the query; and Score-4 (S4), with levels 0 (irrelevant), 1 (related), 2 (partly answering), and 3 (fully answering). Ranking scores are $(2p_{\text{rel}} + p_{\text{part}})/2$, $P(\text{yes})$, and $\mathbb{E}[\ell]/3$, respectively. All three formats share one API call per pair, although Jeff evaluates them in separate forward passes; they are not independent samples. C3 adds $10^{-6}p_{\text{rel}}$ as a secondary score; remaining exact ties use BM25 rank.

Calibration uses probabilities of binarized relevance, distinct from expected-grade ranking scores. Cross-encoder raw probabilities are sigmoid-transformed logits; calibration was not their training objective. Strict C3/S4 readouts are $p_{\text{rel}}/P(\ell \geq 2)$; lenient readouts add partial relevance/$P(\ell \geq 1)$. The lenient policy for TREC-COVID, DBPedia, and News was introduced after observing poor strict C3 calibration. We therefore lead with policy-independent Noul and disclose both Jev C3 results (ECE 14.8 under the policy, 19.2 strictly).

Ranking suite and systems. We rerank identical Pyserini BM25 top-100 candidates from RankGPT/RankLLM [14, 15]: TREC DL19–23 (308 queries) and eight BEIR collections [18] (1,525 queries): TREC-COVID, NFCorpus, Touche, DBPedia, SciFact, Signal, News, and Robust04. The suite has 1,833 queries; 15 NFCorpus topics have empty BM25 candidate lists and score zero. Averages weight the 13 collections equally. Baselines include MiniLM-L6-v2 (22M), bge-reranker-v2-m3 (568M) [2], and Qwen3-Reranker-0.6B/4B/8B [21] (4B/8B use 4-bit NF4). Decision models are Laya (421M), Jeff-Qwen3.5-0.8B v1.2, Clef 27B (seven collections), and closed Jev 1.13 (snapshot `20260917`). Additional controls use Qwen3-235B-A22B's yes/no and expected 0–4 token scores (eight collections), Clef's Qwen3.8-27B base (three), and Jeff's Qwen3.5-0.8B base (all 13), without decision interfaces [22].

Cross-encoders use 512 total tokens; label-probability LLMs read 448 passage tokens. Jev/Clef/Jeff read 4,000 characters and Laya about 320 passage tokens. We additionally compare Jev and bge at 448 passage tokens on Touche, News, and Robust04, and extend bge to 2,048 tokens. Same-base comparisons thus assess configurations with differing prompts and contexts, rather than isolate training effects.

Metrics and inference. We report nDCG@10 and paired query-level t-tests, Holm-corrected across collections for each system–baseline pair. Random-effects pooling uses DerSimonian–Laird heterogeneity estimates, with normal and post-hoc modified Knapp–Hartung (mKH) intervals [13]; mKH uses t_{k-1} and a residual variance multiplier floored at one. The ±2 nDCG equivalence margin was chosen after observing results. Condensed nDCG excludes unjudged passages and cannot establish their relevance.

Calibration uses ECE (15 equal-width bins) and Brier score on judged pairs from nine collections containing negatives: DL19–23,

TREC-COVID, DBPedia, News, Robust04. Relevant grades are ≥ 2 on DL and ≥ 1 on BEIR. Paired query-cluster bootstraps use 2,000 replicates and Holm over nine collections. Platt scaling [11] fits $\sigma(a\,\mathrm{logit}(p) + b)$. Transfer uses source labels only: *All* pools the other collections with equal pair weights; *Group* balances source collections within the same numeric relevance-threshold group (DL or dense BEIR). Grouping does not imply identical relevance semantics. Target-label cross-fitted Platt scaling (10 repeated five-fold query splits) is a reference floor.

RAG protocol. MuSiQue-Ans dev [19] supplies 20 paragraphs with 2–4 supports; NQ-open test [7] uses DPR top-20 passages [6] and string-match labels; 18.6% of final NQ questions lack an answer-bearing candidate. Seed-13 calibration/pilot/final splits contain 200/300/1,000 questions with disjoint IDs. MuSiQue families overlap: 494 final questions reuse their terminal sub-question ID from pilot, 349 from calibration (578 either); 671 share any decomposition ID with either. Calibration labels choose Jev's readout by Brier score (Noul on MuSiQue, strict C3 on NQ) and fit Platt scaling. Post-hoc *raw* thresholds retain this label-selected readout.

Jev ranks by S4 and bge by its logit. Nine policies cover top-1/2/3/5, Adaptive-k (largest score gap) [16], $P > t$ with $t \in \{0.2, 0.3\}$, and top-2 plus further passages with $P > t$, $t \in \{0.3, 0.5\}$. Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and Qwen3-30B-A3B-Instruct-2507 generate short answers through OpenRouter (temperature 0; maximum 32 tokens; no abstention); we score SQuAD-normalized F1. Ranker-effect tests use paired questions, Holm over nine policies per dataset/generator. Threshold-rule tests use a separate four-comparison family. Policies were fixed after pilot; rule comparisons against top-3/Adaptive-k were primary, while top-2/top-5 comparisons and raw thresholds were later exploratory additions. F1 equivalence tests with a post-hoc ±2 margin are uncorrected. Full downstream comparisons cover bge, not Qwen8B.

3 RESULTS

RQ1: Competitive but heterogeneous ranking. Jev S4's average is close to Qwen8B and above bge (Figure 1a). On DL/BEIR, Jev averages 61.1/53.0 versus bge's 60.6/49.3. The pooled Jev–bge effect is +2.6 points: normal 95% CI $[0.1, 5.1]$, mKH $[-0.4, 5.6]$. Jev–Qwen8B is +0.3: normal 90% CI $[-1.3, 1.9]$, mKH $[-1.4, 2.02]$. Thus pooled superiority over bge and ±2 equivalence to Qwen8B do not survive mKH. Heterogeneity is large ($I^2 \approx 0.9$). Collection-level Holm tests favor Jev over bge on five collections and none favor bge; against Qwen8B, Jev wins Robust04 (+7.9) but loses Signal (−3.9). BM25 remains best on Touche. Removing Robust04 reduces Jev/bge averages to 55.3/53.8; the mKH pooled 95% CI still crosses zero ($[-0.4, 3.9]$).

Clef and Jev average 64.4–65.4 and 64.1–65.2 across formats on seven shared collections; their exploratory equivalence survives mKH. Clef's base averages 75.5 on DL19/20 and TREC-COVID versus Clef's 74.8–75.5, with no significant collection-level difference, although mKH does not establish equivalence on only three collections. Jeff exceeds its base (39.1–39.4 vs. 31.8) but remains below BM25 overall. A graded readout improves Qwen235B from 58.1 to 59.8 on its eight collections, versus Jev S4 61.8. These are configuration comparisons.

At equal 448-token passage budgets, Jev exceeds bge on Robust04 (+11.5) and News (+4.9, both Holm $p < 0.01$), and loses on Touche (−9.9). Extending bge largely closes the News gap but not Robust04. Separate newer-content checks yield Jev 62.5–63.5 versus bge 58.2 on FutureQueryEval [1], and 90.4–91.3 versus 81.8 on NovelEval [15]; leads over Qwen8B are not significant. Both sets predate Jev's release, so these checks cannot exclude contamination.

Published monoT5-3B averages 51.4 on BEIR-8 versus Jev's 53.0 on identical candidates [9, 15]; RankZephyr-7B averages 62.1 on DL19–23 versus 61.1 [12, 14]. These are descriptive without matching per-query outputs. Our 4-bit DL19 RankZephyr scores 72.5 versus published 74.1.

RQ2: Raw calibration gains can narrow after transfer. Jev Noul's mean raw ECE is 5.0 points below bge (bootstrap 95% CI $[-6.3, -3.9]$), lower on eight of nine collections and significantly lower on five. Laya improves by 9.2 points; Jeff Noul is worse (Figure 1b). Jev Noul also has lower mean Brier score (0.170 versus 0.215). Group transfer gives bge and Jev similar mean ECE (6.9/6.8), a descriptive comparison without a calibration-equivalence test. Pair-pooled transfer gives bge/Jev ECE 14.9/14.2. Jev and Laya's raw Noul advantage persists when unjudged candidates count as negatives, but calibration remains conditional on the candidate pool and labeling process. Their ECE advantages vary by less than 0.4 points across 5–30 bins and equal-mass binning. On seven shared collections, Clef C3 has raw ECE 7.8 and Brier 0.134 versus Jev C3 ECE 12.5 and bge 20.2; these subset means are separate from Figure 1b.

RQ3: Ranking helps multi-hop RAG; thresholds do not. On the original MuSiQue split, Jev beats bge by 2.6–23.2 F1 across nine policies and three generators (27 question-level Holm tests pass; Table 1, Figure 1c). These tests ignore family dependence. On NQ, only one of 27 passes. Jev S4 puts the final supporting hop first on 51.2% versus bge's 15.2%, despite lower overall support hit@1 (82.2% vs. 88.0%). This association does not identify a cause. Gold-answer masking inserts an answer-location cue: 999/1,000 final-hop paragraphs carry the marker versus 1,621/19,988 candidates; it cannot isolate reasoning from memorization.

Retrospective sensitivity. The 329 questions with no decomposition ID directly shared with calibration or pilot give top-3 gains of +2.4/+5.5/+5.7 F1 (Q2/L3/Q3); none passes nine-policy question-level Holm. Paired family-bootstrap 95% CIs are $[-2.6, 7.3]/[1.0, 10.0]/[0.8, 10.5]$, unadjusted (20,000 resamples, seed 13; 212 subset-bearing families, defined across all 2,417 source questions by shared IDs; question-weighted means). Existing fits, policies, and answers stay fixed. Jev's point gains remain positive across all nine policies. Floor-2+$P > 0.3$ versus top-3 or Adaptive-k has family CIs crossing zero for all generators. Case mix changes, and 23 questions remain indirectly connected to design families: this is neither fully family-disjoint nor confirmatory.

On the original splits, Platt $P > 0.3$ leaves 10.3%/21.6% of MuSiQue/NQ questions without context. Top-2-plus-threshold selects 3.0/3.5 passages, saving none over top-3, with no significant fixed-k gain. Post-hoc ±2 F1 equivalence holds on NQ across generators; MuSiQue is equivalent to top-2 or slightly worse than larger/adaptive selections. MuSiQue raw $P > 0.3$ leaves 25%

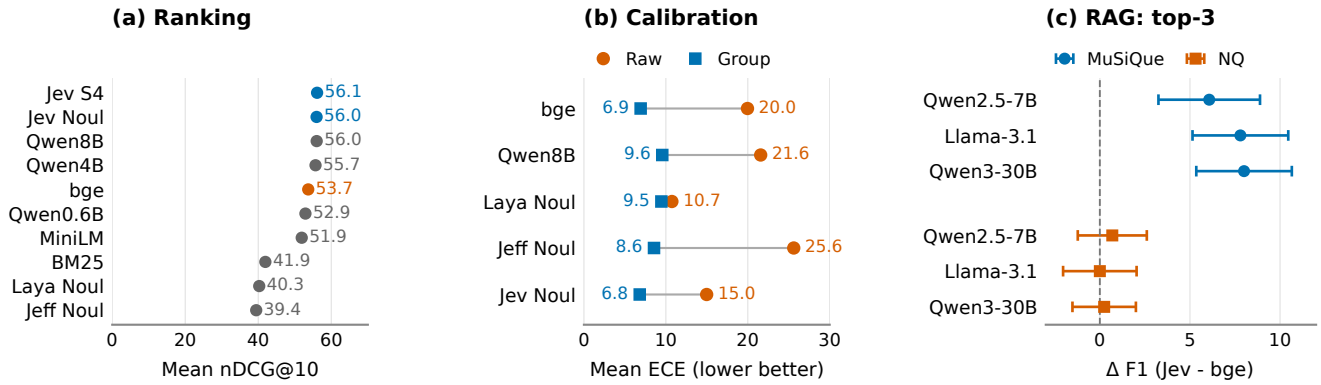


Figure 1: (a) Equal-weight nDCG@10 over 13 collections. (b) Raw versus numeric-group transfer ECE on nine judged pools; decision readouts are Noul. Means are descriptive, in points. (c) Top-3 Jev–bge F1 differences on the original 1,000-question splits, with ordinary paired 95% CIs and question-level Holm tests; MuSiQue family dependence is unaccounted for. Qwen denotes Qwen3-Reranker in (a,b).

Table 1: Answer F1 on the original 1,000-question splits, plus the retrospective MuSiQue subset excluding directly shared components. Q2/L3/Q3: Qwen2.5-7B, Llama-3.1-8B, Qwen3-30B-A3B. *n*: mean passages. Original MuSiQue pairs pass question-level Holm tests, which ignore family dependence; clean top-3 pairs do not. Only NQ L3 top-1 passes. Thresholds use target-label Platt scaling.

		MuSiQue				NQ			
Ranker	Policy	Q2	L3	Q3	*n*	Q2	L3	Q3	*n*
–	All 20	38.8	37.0	46.3	20.0	52.5	49.4	51.5	20.0
–	Gold (oracle)	60.9	54.7	64.1	2.6	60.2	59.1	59.7	3.4
bge	Top-1	22.0	18.0	23.7	1.0	47.9	47.1	49.1	1.0
bge	Top-3	38.5	31.5	39.1	3.0	51.4	50.5	52.6	3.0
bge	Adaptive-k	30.5	26.1	32.6	3.1	50.6	50.5	51.8	7.1
bge	Top-2+$P > 0.3$	35.7	30.4	37.9	2.9	51.3	51.2	52.6	3.9
Jev	Top-1	45.2	36.4	43.4	1.0	50.4	50.4	51.4	1.0
Jev	Top-3	44.6	39.3	47.1	3.0	52.1	50.5	52.9	3.0
Jev	Adaptive-k	46.7	40.2	47.2	2.9	52.3	50.1	52.3	6.2
Jev	$P > 0.3$	43.6	36.1	43.5	2.4	51.2	51.5	52.3	2.9
Jev	Top-2+$P > 0.3$	44.2	39.1	45.2	3.0	51.8	51.0	53.2	3.5
MuSiQue subset: 329 questions with no directly shared decomposition ID									
bge	Top-3 clean	41.3	35.7	42.4	3.0	–	–	–	–
Jev	Top-3 clean	43.6	41.1	48.1	3.0	–	–	–	–

empty and trails top-3 by 3.4–5.9 F1; its readout uses calibration labels. The gold-context gap remains substantial. Jev's advantage is chiefly ranking, rather than calibrated thresholding.

4 DISCUSSION AND CONCLUSION

Typed decision models can be competitive passage rerankers, but neither their interface nor lower raw ECE guarantees improved context selection. Strong trained rerankers remain essential comparisons; the full RAG study establishes no advantage over Qwen8B. Same-base controls do not isolate decision training, and long-document gains vary by collection.

Efficiency. On an 8 GB RTX 4060, median pair latency is 0.3–14 ms for small trained rerankers and 102/148 ms for 4-bit Qwen4B/8B; Jev takes 176 ms per concurrent API call. Serving differences prevent controlled speed comparisons. Jev costs $3.4 per 1,000 depth-100 queries; all API experiments cost about $31. Generator savings exclude candidate scoring.

Limits and reproducibility. BM25 candidates, one prompt per format, incomplete qrels, propagated DL22 judgments, and 31 synthetic DL23 queries constrain interpretation. Jev's size, training data, and cutoff are unknown; contamination remains possible. Generator quantization may vary by provider; historical weight revisions were not recorded. Cached runs, input hashes, split IDs, and scripts support numerical verification. MuSiQue calibration/pilot component reuse and retrospective analyses limit inference. Future work needs family-disjoint splits, matched contexts, and a downstream Qwen8B baseline.

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